

CLAUDE.md

Project: Humanitarian Antenna Mast Field Station

This repository supports the design, engineering, documentation, and software for a humanitarian/off-grid antenna mast and field station at a private home test site.

The project owner works in international humanitarian information management and digital response, including Digital Surge Officer, IT/ITT Officer, and Information Management Officer roles. This site is intended to serve as a practical home test bed for piloting small communications, sensing, edge-compute, lighting, power, and field-networking devices before they are considered for humanitarian field deployment.

The project is not just a hobby antenna installation. It is a disciplined humanitarian communications and field-systems lab that should teach, document, and validate patterns that could be replicated under constrained field deployment conditions.

Core Goal

By **November 15, 2026**, complete a safe, grounded, weather-ready humanitarian mast station operating at the private test site.

The first successful build must include:

- A 50-foot H50 / 25G-style mast concept.
- A top-mounted VHF/UHF antenna.
- One HF wire system.
- Bonded coax and surge protection back to the shack.
- A weatherproof field enclosure at the mast base.
- A Raspberry Pi-based field station.
- Lighting control for mast/tree/site lighting.
- A low-power DC architecture that can be practically replicated off-grid.
- Documentation sufficient to rebuild, inspect, and troubleshoot the system.

Humanitarian Operating Context

This mast and field station should be treated as a practical test bed for humanitarian field systems, not merely a home antenna project.

The station should help pilot, validate, document, and harden small deployable devices before they are considered for field deployment.

Relevant humanitarian use cases include:

- Emergency communications experimentation.

- Field connectivity patterns.
- Low-power edge computing.
- Environmental and situational sensing.
- BirdNET / acoustic monitoring.
- Lighting and remote control systems.
- Solar-capable DC infrastructure.
- Raspberry Pi-based field services.
- Repeatable deployment kits.
- Training and documentation for field teams.

The standard for success is not only “works at home.” The standard is “could be understood, packed, rebuilt, powered, troubleshoot, and adapted by a small field team under constrained conditions.”

Field-Deployment Design Discipline

When designing hardware, software, documentation, or operating procedures for this repo, assume the project may later inform humanitarian field deployment kits.

Prefer designs that are:

- Low-power.
- Solar-capable.
- DC-native where practical.
- Rugged.
- Modular.
- Easily documented.
- Easy to inspect.
- Easy to repair.
- Understandable by non-specialists.
- Rebuildable from commonly available parts.
- Able to recover after power loss.
- Not dependent on cloud services for core operation.
- Not dependent on home-only conveniences.

Avoid designs that:

- Work only because they are installed at a comfortable home site.
- Require fragile configuration knowledge.
- Depend on undocumented manual steps.
- Assume unlimited AC power.
- Assume stable internet.
- Require specialized tools without justification.
- Add complexity before the baseline system is stable.
- Would be embarrassing or impractical to recommend for field replication.

Critical Milestones

July 2026 — Scope Freeze and Site Plan

Complete:

- Mast siting concept.
- Guy/anchor geometry.
- Field enclosure placement.
- Conduit/fiber route.
- Grounding and bonding plan.
- Initial antenna stack.
- First-build bill of materials.
- Permit / utility / inspection questions list.

August 2026 — Civil and Utility Work

Complete or prepare:

- Mast base.
- Guy anchors.
- Grounding electrode and bonding materials.
- Conduit sleeves or trenching.
- Fiber/data path.
- Shack entry panel / bulkhead concept.
- Field enclosure mounting point.

September 2026 — Mast and RF Build

Critical checkpoint:

- By **September 30, 2026**, the mast should be standing and bonded.

Complete:

- Mast assembly.
- Guy installation and tensioning.
- Bonding and grounding.
- VHF/UHF vertical.
- HF wire support.
- Coax path to shack.
- Surge protection.
- Basic continuity and RF checks.

After September 30, avoid heavy construction scope. The work should shift to tuning, cleanup, software, monitoring, documentation, and commissioning.

October–Mid-November 2026 — Commissioning

Complete:

- Raspberry Pi field station.
- Lighting controller.
- BirdNET-ready field service capability.
- Basic telemetry.
- Remote health monitoring.
- Winter inspection checklist.
- As-built drawings.
- Operating notes.
- Shutdown / storm procedure.

Settled Site Concept

Use the accepted private site layout from the private geospatial/source-of-truth repository.

Public documentation must describe this only as a **private test site** or **representative rural field-station test bed**.

Private planning assumptions are maintained separately and may include exact mast placement, real parcel geometry, real guy-anchor layout, real conduit route, and real field enclosure placement.

Important: public documents must not include the street address, parcel number, real coordinates, real property geometry, map screenshots, KML/KMZ/GPX/GeoJSON artifacts, or identifiable aerial imagery.

Product Discipline

The first build must stay disciplined.

The system should start as a **passive mast-and-utility platform with solar-capable edge services**, not as a complex remote radio site.

Prioritize:

1. Safe mast.
2. Good grounding and bonding.
3. Clean coax return to shack.
4. Fiber/data path.
5. Weatherproof field enclosure.
6. Low-power DC services.
7. HF and VHF capability.
8. Lighting control.
9. Documentation.

Do not allow cameras, rotators, Yagis, LoRa, LTE, APRS, airband, remote transmit radios, or solar/battery expansion to derail the first build.

First-Build Scope

The first iteration includes:

- 50-foot mast concept.
- Mast base.
- Guy anchors.
- Bonding and grounding.
- Coax surge protection.
- Shack entry/bulkhead.
- Top-mounted VHF/UHF vertical.
- One HF wire antenna system.
- Field enclosure.
- Raspberry Pi.
- Lighting controller.
- Low-power DC architecture.
- Fiber or fiber-ready data path.
- Basic system documentation.

The first iteration does **not** include:

- Rotator.
- Beam/Yagi.
- Multiple experimental RF services.
- Remote transmit radio.
- Remote tuner unless later justified.
- Camera sprawl.
- Permanent LTE dependency.
- High-power AC loads.
- Non-field-replicable convenience features.

Antenna Priorities

The first RF priority is **HF + VHF/UHF**.

VHF/UHF

Use one clean, serious, top-mounted dual-band VHF/UHF vertical.

Design expectations:

- Mount at or near mast top.
- Keep the top uncluttered.
- Coax returns to the shack.

- Include lightning/surge protection.
- Bond at the appropriate entry/bulkhead.
- Avoid early addition of side whips or unrelated services.

HF

Use the mast as the high support for one HF wire system.

Acceptable first designs:

- EFHW.
- Inverted-L.
- Sloper.
- Similar simple wire antenna.

Design expectations:

- Feedpoint, transformer, arrestor, drip loop, and counterpoise/radial plan should be at or near the mast base unless a better reason is documented.
- Coax should return to the shack through bonded protection.
- Keep tuning and maintenance simple.
- Favor field-replicable patterns over maximum performance complexity.
- Public documents should show generalized geometry, not real site geometry.

Field Enclosure

The field enclosure is part of the first build.

Its purpose is edge services, not full remote radio operation.

Initial services:

- Raspberry Pi field station.
- Lighting controller.
- BirdNET-ready microphone support.
- Environmental sensors.
- DC power distribution.
- Fiber/data termination.
- Basic telemetry.
- Possible receive-only SDR later.

Avoid placing primary transmit radio functions in the field enclosure during the first build.

Off-Grid / Solar Discipline

All first-build decisions must be compatible with eventual off-grid operation.

Design standard:

- Low-power DC first.
- Solar/battery feasible.
- Field-replicable.
- Maintainable without house-grade infrastructure.
- Minimal idle load.
- Graceful degradation.
- Weather-tolerant.
- Easy to disconnect, inspect, and repair.

Grid power may be used for testing or fallback, but the architecture should not depend on assumptions that could not practically be replicated in an off-grid humanitarian deployment.

When choosing hardware or software, prefer options that:

- Run on 12V/24V DC or efficient DC conversion.
- Recover automatically after power loss.
- Can be monitored remotely.
- Have low idle draw.
- Do not require cloud services for core operation.
- Can be documented and reproduced.

Data and Network Architecture

Preferred data path:

- Fiber between house/shack and field enclosure where practical.
- Avoid copper Ethernet as the primary long outdoor data path because it creates an additional conductive lightning pathway.
- Keep the field box network simple and robust.
- Assume outages and power cycling will happen.

The Pi should support:

- Local control.
- Local logging.
- Remote access over the site network.
- Recovery after reboot.
- Clear service documentation.

Grounding, Bonding, and Safety

Safety and bonding are first-class requirements.

Do not generate instructions that imply this is a casual temporary mast.

Any design work should assume:

- Proper mast grounding.
- Proper coax surge protection.
- Bonded entry panel.
- Avoidance of isolated ground islands.
- Coordination with the premises grounding electrode system.
- Utility locate before digging.
- Permit/code review where applicable.
- Structural review where required.
- Conservative weather assumptions.

Do not treat public GIS, conceptual drawings, or sanitized diagrams as construction-grade survey information.

Any staking, trenching, concrete, guy anchor installation, or permanent tower work must be field-verified.

Public / Private Geospatial Versioning Strategy

This project will produce geospatial and site-derived engineering data products that must be preserved, versioned, backed up, and reproducible.

The strategy is not to ignore or discard sensitive geospatial work. The strategy is to separate **private authoritative geospatial artifacts** from **public sanitized engineering artifacts**.

Core Principle

Maintain two classes of project artifacts:

1. Private source-of-truth artifacts

2. Real site plans.
3. Real coordinates.
4. Real parcel geometry.
5. Real mast/guy/enclosure/conduit layouts.
6. Real KML/KMZ/GeoJSON/GPX/QGIS files.
7. Real survey-derived or GIS-derived products.
8. Photos or maps that reveal the home site.

9. Public sanitized derivatives

10. Generalized engineering diagrams.
11. Local-relative coordinate examples.
12. Abstracted site layouts.

13. Non-georeferenced mast/guy concepts.
14. Reusable patterns for humanitarian field deployment.
15. BOMs, code, procedures, checklists, and architecture notes.

Private artifacts must be versioned and backed up. Public artifacts must be safe to share.

Repository Model

Use a split repository model.

```
humanitarian-mast-public/  
├─ CLAUDE.md  
├─ README.md  
├─ docs/  
├─ diagrams/  
├─ bom/  
├─ pi-field-station/  
├─ lighting-controller/  
└─ checklists/  
  
humanitarian-mast-private/  
├─ README.md  
├─ CLAUDE.md  
├─ geospatial/  
│   ├── source/  
│   ├── processed/  
│   ├── qgis/  
│   ├── kml/  
│   ├── geojson/  
│   ├── rasters/  
│   └─ exports/  
├─ site-engineering/  
│   ├── mast-location/  
│   ├── guy-anchors/  
│   ├── conduit-routes/  
│   ├── grounding/  
│   └─ shack-entry/  
├─ photos/  
├─ permits/  
├─ manifests/  
└─ sanitized-exports/
```

The private repository is the authoritative engineering record for real site work.

The public repository contains only sanitized, generalized, or synthetic products.

Private Versioning Requirements

The private repository should be versioned and backed up with the same seriousness as code.

Private artifacts should include:

- Raw GIS exports.
- QGIS projects.
- KML/KMZ files.
- GeoJSON files.
- Survey images or PDFs.
- Field-staked coordinates.
- Mast and guy-anchor layout files.
- Utility route sketches.
- Grounding layout drawings.
- Site photos.
- Construction photos.
- As-built drawings.
- Derived engineering drawings.
- Metadata and processing notes.

Do not rely on a single local workstation, NAS, or server.

Use at least:

- One local working copy.
- One private remote Git repository.
- One additional backup path.

For large files, use Git LFS, DVC, git-annex, restic, or another explicit artifact/versioning strategy rather than leaving files outside version control.

Private Geospatial Data Layout

```
geospatial/  
├── source/  
│   ├── county-gis/  
│   ├── survey/  
│   ├── parcel/  
│   ├── aerial/  
│   └── field-measurements/  
├── processed/  
│   ├── local-coordinate-models/  
│   ├── mast-layouts/  
│   ├── guy-anchor-layouts/  
│   └── conduit-routes/
```

```
|   └─ grounding-layouts/
├─ qgis/
|   └─ project.qgz
|   └─ styles/
├─ kml/
├─ geojson/
├─ rasters/
├─ exports/
└─ README.md
```

Every private geospatial product should have enough notes to explain:

- Source.
- Date acquired.
- Processing steps.
- Coordinate reference system.
- Accuracy limitations.
- Whether it is conceptual, field-measured, survey-derived, or as-built.
- Whether it is safe or unsafe for public release.

Public Sanitization Pipeline

Public diagrams and data products should be generated from private source material through an intentional sanitization step.

Public derivatives may use:

- Local-relative coordinates.
- Synthetic example coordinates.
- Simplified mast/guy geometry.
- Abstract site outlines.
- Non-identifiable rectangular test-site diagrams.
- General direction labels.
- Approximate engineering distances when they cannot reveal the property.
- Placeholder names such as `private-test-site`.

Public derivatives must not include:

- Street address.
- Parcel identifier.
- Latitude/longitude.
- Real KML/KMZ/GPX/GeoJSON.
- Real parcel boundary geometry.
- County GIS screenshots.
- Google Maps screenshots.
- Aerial imagery.
- Survey images.

- Utility routes tied to the real property.
- Exact mast or anchor coordinates.
- Road names.
- House numbers.
- EXIF GPS metadata.

Sanitized Export Directory

The private repository may contain a `sanitized-exports/` directory that stages public-safe artifacts before copying them into the public repository.

```
sanitized-exports/
├── diagrams/
│   ├── conceptual-mast-layout.svg
│   ├── generic-guy-anchor-layout.svg
│   ├── field-enclosure-architecture.svg
│   └── grounding-pattern.svg
├── docs/
│   ├── antenna-design-public.md
│   ├── solar-dc-public.md
│   └── commissioning-public.md
└── manifest-publication-review.md
```

No file should move from `sanitized-exports/` to the public repository until it passes privacy review.

Data Product Classification

Every generated artifact should be classified.

Use these labels:

```
classification:
  visibility: private | public-safe | review-required
  geospatial_sensitivity: none | low | medium | high
  contains_real_coordinates: true | false
  contains_real_parcel_geometry: true | false
  contains_identifiable_imagery: true | false
  safe_for_public_repo: true | false
```

If an artifact contains real coordinates, real parcel geometry, identifiable imagery, or address-derived information, it is private by default.

Manifest and Checksums

The private repo should include manifests for important artifacts.

Example:

```
artifact:
  name: mast-layout-private-v1.geojson
  type: geospatial
  visibility: private
  source: private GIS and planning geometry
  date_created: 2026-06-30
  coordinate_system: real-world CRS
  checksum: "<sha256>"
  public_derivative: conceptual-mast-layout.svg
  notes: "Contains real site geometry. Do not publish."
```

Use checksums for important generated outputs so that engineering products can be verified later.

Recommended Git Ignore Strategy

Do not globally ignore all geospatial data.

Instead:

- The **public repo** should ignore geospatial and private artifacts.
- The **private repo** should version geospatial artifacts intentionally.
- Large private artifacts should be handled with Git LFS, DVC, git-annex, or another durable artifact store.
- Temporary scratch exports may be ignored.
- Final private source and derived products should be tracked or explicitly backed up.

Public repo `.gitignore` should be strict:

```
# Never publish private site data
/private/
/docs/private/
/diagrams/private/
/geospatial/
/site-engineering/

# Never publish real geospatial artifacts
*.kml
*.kmz
*.gpx
```

```
*.geojson
*.shp
*.shx
*.dbf
*.prj
*.qgz
*.qgs

# Avoid accidental location leakage
*parcel*
*survey*
*site-map*
*property-boundary*
*aerial*
*satellite*
*plat*
*csm*
*address*
```

Private repo `.gitignore` should ignore only scratch and temporary files:

```
# Temporary GIS and render files
*.tmp
*.bak
*.aux.xml
.qgis2/
__pycache__/

# Scratch exports
scratch/
tmp/
working/
```

Photo and Media Rules

Before publishing any photo:

- Strip EXIF metadata.
- Remove GPS metadata.
- Check background landmarks.
- Avoid showing road frontage, mailbox, house number, driveway layout, or nearby unique buildings.
- Avoid showing screenshots that include map UI, parcel outlines, or recognizable aerial context.
- Prefer close-up construction photos over wide site-context photos.

Use sanitized filenames. Do not include address, parcel, road name, county GIS identifiers, or other location clues in filenames.

Pre-Publication Review Gate

Before copying any artifact from the private repo to the public repo, perform a review.

Checklist:

- Does it include the real address?
- Does it include a parcel number?
- Does it include real coordinates?
- Does it include real parcel geometry?
- Does it include a real KML/KMZ/GPX/GeoJSON file?
- Does it include county GIS, Google Maps, or aerial imagery?
- Does it include road names or nearby landmarks?
- Does it include EXIF GPS metadata?
- Does the filename reveal the location?
- Could a motivated reader identify the home site from this artifact?

If yes, do not publish. Create a sanitized derivative instead.

Engineering Guardrails

When helping with this repo, Claude should:

- Preserve the settled scope unless explicitly changed.
- Warn when a proposed feature increases complexity before the baseline is stable.
- Separate “first build,” “later enhancement,” and “bad idea for now.”
- Prefer durable civil/electrical infrastructure over buying radios early.
- Prefer simple, inspectable, field-replicable systems.
- Create checklists, diagrams, BOMs, scripts, and commissioning notes.
- Call out safety, permitting, grounding, and structural concerns clearly.
- Avoid pretending conceptual site data is survey-grade.
- Avoid over-optimizing antennas before the mast/grounding path is solved.
- Keep documentation practical enough to use outdoors during construction.

Recommended Public Repository Structure

```
.
├── CLAUDE.md
├── README.md
├── docs/
│   ├── goals.md
│   ├── public-architecture.md
│   ├── milestones.md
│   ├── safety-grounding-bonding.md
│   ├── antenna-plan.md
│   └── field-enclosure.md
```

```

|   |─ solar-dc-power.md
|   |─ commissioning.md
|   └─ winterization.md
|─ diagrams/
|   |─ public/
|   |   |─ conceptual-layout.svg
|   |   |─ power-architecture.svg
|   |   |─ network-architecture.svg
|   |   └─ grounding-pattern.svg
|   └─ templates/
|─ bom/
|   |─ first-build-bom.md
|   |─ deferred-items.md
|   └─ vendor-notes.md
|─ pi-field-station/
|   |─ README.md
|   |─ services/
|   |─ scripts/
|   |─ config/
|   └─ systemd/
|─ lighting-controller/
|   |─ README.md
|   |─ wiring.md
|   |─ controller-config.md
|   └─ test-plan.md
|─ checklists/
|   |─ pre-dig.md
|   |─ mast-raise.md
|   |─ grounding-inspection.md
|   |─ rf-test.md
|   |─ field-box-test.md
|   |─ publication-review.md
|   └─ storm-shutdown.md
|─ archive/
|   └─ prior-concepts/

```

Documentation Standards

All project documents should distinguish between:

- Settled decisions.
- Working assumptions.
- Open questions.
- Deferred features.
- Safety requirements.

- Field verification tasks.
- Public-safe artifacts.
- Private artifacts.
- Review-required artifacts.

Use concise engineering language.

Prefer tables and checklists where useful.

Every BOM should mark items as:

- Required for first build.
- Nice to have.
- Deferred.
- Do not buy yet.

Every data product should mark whether it is:

- Public-safe.
- Private.
- Review-required.

Acceptance Criteria for First Build

The first build is successful when:

- Mast is standing, guyed, and bonded.
- VHF/UHF vertical is installed and tested.
- HF wire is installed, protected, and tested.
- Coax runs are protected and enter the shack through a bonded path.
- Field enclosure is mounted, weather-protected, and serviceable.
- Pi boots reliably and restarts services after power loss.
- Lighting controller works locally and remotely.
- The system can run in a low-power DC mode.
- Documentation includes as-built notes, diagrams, BOM, and inspection checklists.
- Private geospatial data products are versioned and backed up.
- Public-facing artifacts are sanitized.
- Deferred items remain deferred.

Deferred Enhancement Backlog

Do not implement these until the first-build baseline is stable:

- APRS.
- Airband.
- LoRa.
- LTE failover.

- Cameras.
- Rotator.
- Yagi/beam.
- Remote transmit radio.
- Remote tuner.
- Solar/battery expansion beyond what is needed to validate the DC architecture.
- Multiple SDR services.
- Weather station expansion.
- Permanent automation dashboard.

Claude Behavior for Public Artifacts

When asked to generate public-facing project materials, Claude should:

- Remove exact address references.
- Remove parcel references.
- Replace real site diagrams with sanitized conceptual diagrams.
- Avoid publishing real KML/KMZ/GPX/GeoJSON.
- Avoid map screenshots.
- Avoid county GIS references.
- Use local-relative or synthetic coordinates.
- Mark public examples as representative, not site-specific.
- Ask whether an artifact is intended for public or private use when unclear.
- Default to privacy-preserving output for anything likely to be published.

Claude Behavior for Private Geospatial Work

When working in the private repo, Claude may help produce real geospatial engineering artifacts, but must mark them private.

When working in the public repo, Claude must not include real geospatial details.

When unclear whether an artifact is public or private, Claude should default to `review-required`.

Claude should help create:

- Private source-of-truth layouts.
- Sanitized public derivatives.
- Manifests.
- Checksums.
- Publication review checklists.
- Scripts that strip or transform sensitive geospatial detail.
- Clear separation between public and private outputs.

Claude must not suggest that important private geospatial artifacts remain only on a local machine.

Preferred Claude Behavior

When asked to help:

- Start from this file.
- Confirm whether the request affects first-build scope.
- Keep the project aligned to the November 15, 2026 goal.
- Protect the September 30, 2026 mast-standing-and-bonded milestone.
- Produce practical artifacts that can be saved directly into the repo.
- Flag anything that should be reviewed by a tower professional, electrician, or local authority.
- Do not invent dimensions, code requirements, parcel facts, equipment specifications, or geospatial details.
- When uncertain, label assumptions clearly and create a verification task.
- Treat private geospatial artifacts as durable engineering records, not disposable scratch data.
- Treat public artifacts as sanitized derivatives, not copies of private site products.

Project Rule

Private geospatial artifacts are not disposable.

They must be versioned, backed up, and reproducible.

Public artifacts must be safe, generalized, and location-protective.

The first build should prove the core pattern: **HF + VHF, safe mast, bonded entry, solar-capable field enclosure, Pi edge services, lighting control, and field-replicable documentation.**